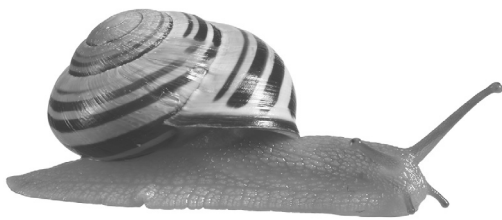
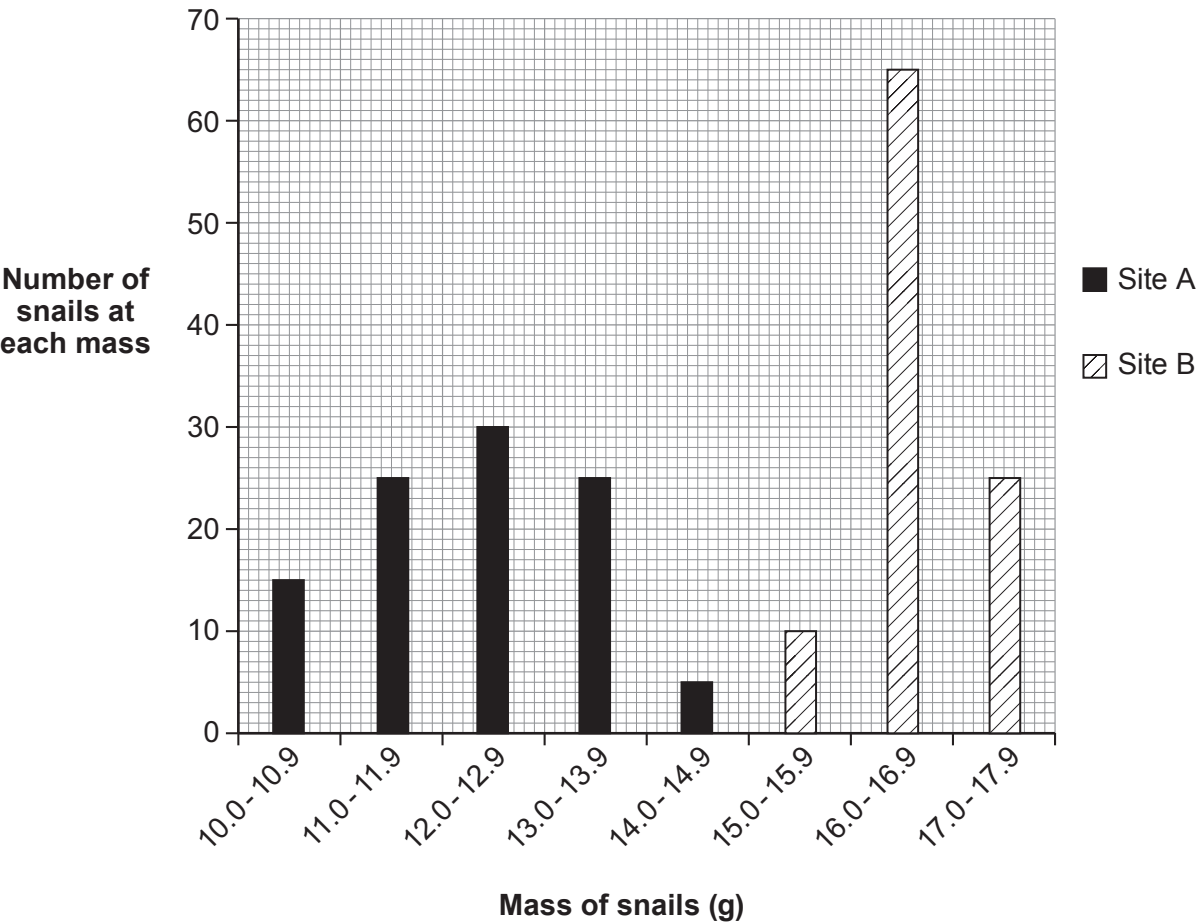


8. The photograph shows the banded snail, *Cepaea nemoralis*.



(a) Scientists investigated variation in the mass of individual snails sampled at random from two different sites, **A** and **B**. The mass of each snail was recorded to the nearest 0.1 g. The results are shown in the bar chart.



Examiner
only

(i) The table below shows the mean mass of snails collected at each site.

Site	Mean mass (g)
A	12.3
B	16.8

Calculate the percentage increase in the mean mass of the snails at site **B** compared to site **A**.

[2]

increase in mean mass = %

(ii) At which of the two sites do the snails show the greater variation in mass? Give the reason for your choice. [1]

.....

.....

(iii) How did the scientists reduce bias in their investigation? [1]

.....

(iv) Why is it important that other scientists carry out the same investigation as these scientists? [1]

.....

(b) *Cepaea nemoralis* shows genetic variation.

Use your knowledge of natural selection to explain the long term advantage of genetic variation to *Cepaea nemoralis* in a changing environment. [3]

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END OF PAPER

